[Original text]

Early Polychrome Glazes on a Chinese Ceramic Bead of the Warring States Period

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Introduction

Chinese glass 'eye beads' from early in the Warring States period (480-221 BC) are believed to represent some of the first glass artefacts made in China (Lu and Hu 1988, An 1996). Their style is thought to derive from imported western beads with similar ornament, but they are considered Chinese because of their characteristic lead oxide-barium oxide-silica compositions, which are rare outside the country (Brill et. al. 1991b, An 1996). However, in parallel with this early Chinese glass tradition can also be found some unusual Chinese glazed ceramic beads made in the 'eye bead' style. These have hard faience-type bodies and often bear superimposed coloured glazes, arranged in patterns imitating those used on eye beads made entirely from glass.

Faience is relatively rare in China, but the polychrome glazed beads, which are the subject of the present paper, have a very faience-like appearance. Their bodies consist of a hard cream-white material covered with an opaque white glaze. Applied to this white glaze-ground are red-brown, opaque-yellow and cold blue glazes, often superimposed in formal geometric designs that often stand in high relief.

Monochrome faience beads with low-fired glazes have been found in China in early Western Zhou (1027-771 BC) contexts, but have generally been considered imports from the West (Zhang et al 1986). Although Brill, Tong and Zhang (1989) found the interstitial glass of one such bead to be rich in potash, and perhaps consistent with a Chinese origin, a later publication (Brill et al. 1991a) considered the origin of the bead open. Thus it is possible that the type of beads discussed here represent the earliest low-fired glaze technology in China. It is also highly probable that they bear examples of the earliest low-fired polychrome glazes used in the country.

Fig. 1 Chinese Warring States faience bead with red-brown, yellow, blue and white glazes on a white glazed ground. L. 2 cm analysed in the present study.

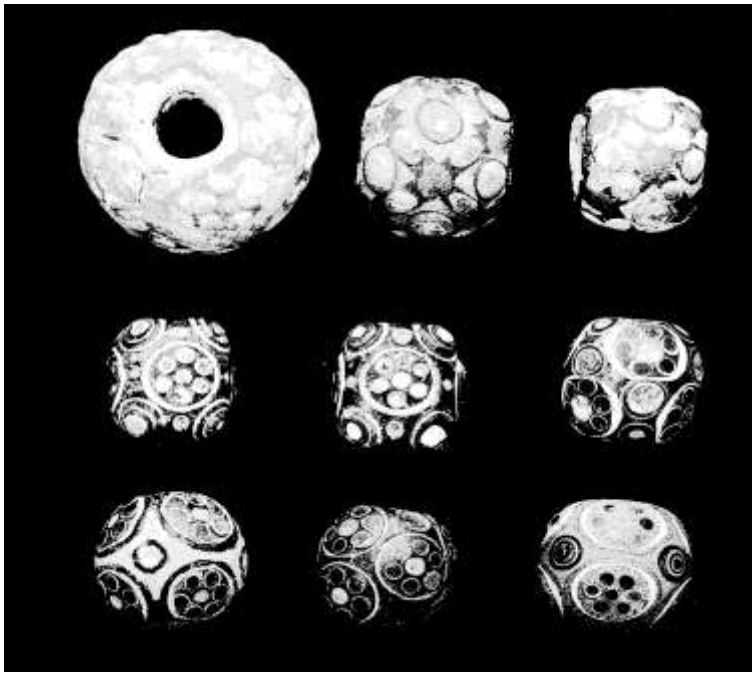

Fig. 2, Typical Chinese Warring States faience eye beads (top row) and glass eye beads (second and third rows). (After Yoshimizu, 1989).

While many examples of Chinese glass beads have been analysed their relationships with contemporary glazed ceramic beads are unclear. In order to explore the technology used to

make these early Chinese ceramic beads, and to understand their connections with contemporary Chinese glass, we have analysed a single Chinese polychrome glazed bead of the Warring States period. This bead is of interest because both stylistically and technologically its glazes appear closely related to those on an earthenware jar of the Warring States period, reported on by Wood and Freestone 1995. The main techniques used for the present study were scanning electron microscopy and energy dispersive X-ray analysis (SEM), with X-ray diffraction (XRD) used to confirm the identity of some minerals.

Results

The bead-body. In the SEM, the pale stony body of the bead proved to have a highly heterogeneous structure, composed of numerous glass-rich fragments which were sintered together but had not fully fused and flowed, leaving a high proportion of open pores. These glass particles are very variable in composition and in some cases have partially devitrified to crystallites of diopside, feldspar or apatite. The bulk analysis shows about 70% silica – a figure typical of many clays or glasses, but much lower than faience from Western Asia. However, the body is neither rich in alumina, like a clay, nor rich in alkalis, like a glass; instead it is rich in alkaline earths and phosphate:

Table 1 – Analysis of bulk body of bead

	SiO_2	Al_2O_3	Fe_2O_3	CaO	MgO	K_2O	Na_2O	P_2O_5	MnO
Bulk bead body	69.4	3.5	1.4	7.8	5.7	5.0	1.8	4.7	0.3

Thus, although superficially similar in appearance to the faience of the West, the body of the bead is not composed largely of quartz, which is an essential component of faience as normally defined.

The bead glazes. The glazes on the bead contained many mineral inclusions (often quartz and albite), suspended in lead oxide- and baria-rich glasses. The angular particle shapes, coupled with reaction rims on the feldspars, indicate that they are firing relicts of glaze-batch material, rather than the products of devitrification. The compositions of the four glaze-colours proved as follows: --

Table 2 – White, blue, yellow and red glazes used on bead

	SiO_2	TiO_2	Al_2O_3	Fe_2O_3	CuO	CaO	MgO	K_2O	Na_2O	BaO	PbO	Cl
White	58.4	0.2	4.8	0.5	<0.1	0.7	0.4	1.0	2.6	11.2	19.1	0.8
Blue	49.9	0.9	4.4	0.5	2.4	0.9	0.4	1.2	2.2	14.8	21.9	0.5
Yellow	27.0	0.4	8.6	4.7	<0.1	1.5	0.7	0.7	0.8	20.1	35.0	0.5
Red	42.3	<0.1	2.0	17.8	<0.1	2.2	0.4	0.7	1.1	8.9	24.2	0.5

In terms of their microstructures, the various glazes showed the following features: -

The opaque white glaze. Abundant angular fragments of quartz, up to about 0.2-mm diameter, opacify this glaze together with a subordinate amount of albite of about the same size. The glassy matrix has not reached equilibrium and the distribution of lead is uneven. Low levels of Fe_2O_3, CaO, and MgO suggest that these are impurities associated with the major ingredients, rather than conscious additions.

Fig. 3 SEM back-scattered image showing amorphous vesicular bead-body and white glaze containing angular quartz and albite fragments in a high Pb+Ba glass.

The opaque blue glaze. This glaze also contained many partially reacted quartz and albite grains. Additionally small crystals, less than 0.005 mm long, were identified as barium copper silicate, containing about 31% BaO, 15% CuO and 51% SiO_2, very close to the ideal composition of the rare mineral effenbergerite ($BaCuSi_4O_{10}$). The presence of this mineral was confirmed by XRD analysis. The morphology of the crystals suggests that they grew from the glaze mixture and are therefore related to the synthetic version of this mineral, Chinese blue.[1]

The opaque red-brown glaze. Quartz grains are present in the red-brown glaze, but not as abundantly as in the white. The reddish colour of the glaze, and much of its opacity, comes from angular grains of hematite, up to about 0.07-mm diameter. No feldspar inclusions were found, and this is reflected in the glaze's lower alumina level (1.96%). The glaze is thinly applied, only up to 0.1 mm in the section taken, compared with 0.3 mm for the yellow.

The opaque yellow glaze. Few grains of quartz are present in this glaze and its opacity is due mainly to abundant, well-formed crystals of the barium feldspar, celsian, $BaAl_2Si_2O_8$ (identified by XRD). The celsian crystals tend to be less than 0.005 mm long. Most of the iron oxide in this glaze (c. 4.6%) is present in the glassy phase, presumably as Fe^{3+}, where it provides a yellow solution colour. Alumina, at 9%, is comparatively high. Lime and magnesia too (at 1.5% and 0.74% respectively) are higher than in the blue and white glazes.

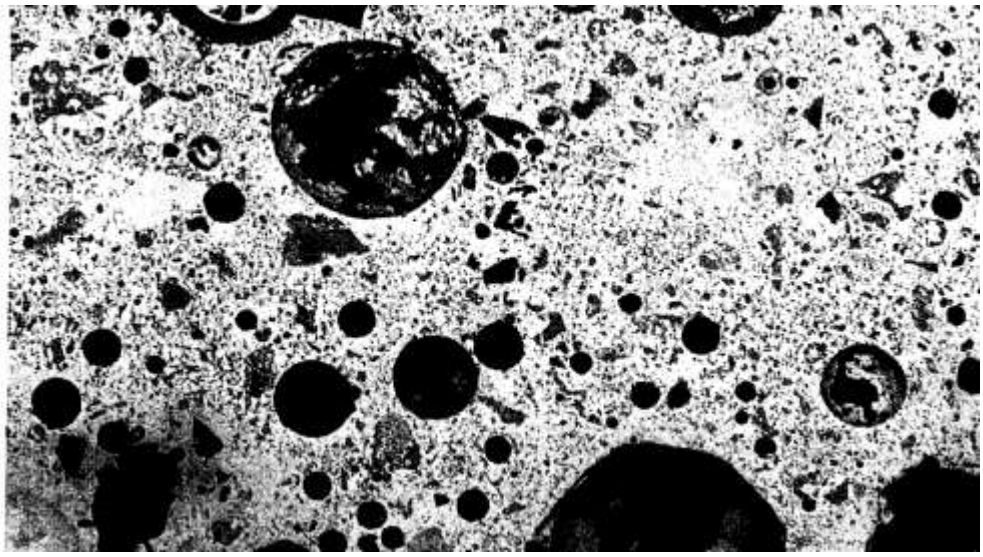

Fig. 4, SEM back-scattered image of yellow glaze showing large vesicles, lack of angular fragments and fine secondary celsian crystals. Same scale as Fig. 3: -- the large vesicle, top centre in Fig. 4, is about 100 mm diameter.

Discussion

The bead body. While the body of the bead seems neither glass nor clay its composition does resemble the ashes of fast-growing plants with tough siliceous stems, such as rice-straw, fern and sorghum: --

Table 3 – Ashes of plants with fast-growing siliceous stems (Ash analyses from Zhang 1986a)

	SiO_2	Al_2O_3	Fe_2O_3	CaO	MgO	K_2O	Na_2O	P_2O_5	MnO
Bulk bead body	69.4	3.5	1.4	7.8	5.7	5.0	1.8	4.7	0.3
Rice-straw ash	75.8	2.1	0.5	5.8	2.6	6.7	1.45	4.7	0.6
Fern ash	70.8	5.5	1.02	8.5	4.0	5.2	0.11	1.2	1.8
Sorghum ash	70.8	5.5	2.5	7.6	3.8	6.0	0.6	1.6	1.8

These parallels raise the possibility that the faience bead body may have been made entirely from botanic ash. The high phosphate content is particularly indicative of a vegetal ash, as is the heterogeneity of the material and its high fusibility. If the body were made from ash, some organic binder or paste may have been used to help form the bead, and to hold the ash together, which may in part explain its high porosity. The use of a ceramic body made entirely from plant ash would have been unusual and, on present evidence, seems to have neither precedents nor descendants in ceramic history. Advantages of this amorphous

material over traditional quartz-based polycrystalline faience compositions (as described in Kaczmarczyk and Hedges, 1983), would have been an expansivity more compatible with lead-baria glazes. The good sintering behaviour of this type of ash would also have supplied a harder and tougher body than faience based upon quartz.

The bead glazes. The coloured surfaces, applied to the bead-body, are true glazes, albeit somewhat underfired. They have not been fritted before application and some of their original batch-material can be seen in the SEM pictures. However, the less-than-complete melting evident in these glazes was probably necessary to preserve the integrity of the applied and superimposed coloured-glaze patterns, and to maintain their opacity and colour. Had these glazes been taken to a full melt they would have merged and run, and lost their definition. This layered approach produced coloured ornament in much higher relief than contemporary Chinese glass eye-beads, where the different coloured glasses were levelled by flame-work and/or marvering. In terms of the glazes' original compositions, the following approaches seem possible:

White. The white glaze was made from crushed quartz and albite mixed with lead and barium minerals. Given the small sample-areas analysed, the BaO/(BaO+PbO) ratios in the white, blue and yellow glazes are relatively constant at 0.36, 0.40 and 0.36, respectively. Naturally occurring lead-barium minerals are apparently unknown but a mineral deposit with a relatively constant lead:barium ratio may have been utilised. The simplest recipe that could have been used for this glaze would have been a mixture of a crushed quartz-feldspar rock, such as a pegmatite or aplite, with a crushed lead-barium ore in proportions of about 2: 1. However, recipes made from separate additions of quartz, feldspar, a lead mineral, and a barium mineral, cannot be ruled out.

Blue. As for white, with an addition of copper. The copper could have been added as oxidised metal, or as a natural copper mineral. However it may have been added in the form of prepared Chinese blue, the tetrasilicate of copper and barium, prepared with a small amount of lead oxide and which, along with its related purple variant, was often used as a paint pigment in Warring States and Han China (FitzHugh and Zycherman, 1983, 1992, Herm et al 1995a, Herm et al 1995b). Although the Chinese blue found in the glaze seems to have crystallised during firing, it is not impossible that these crystals developed and grew from dissolved Chinese blue additions. This possibility is supported by the slightly higher ratio of barium: lead in the blue glaze, relative to the yellow and the white (see above), suggesting that a barium-rich but low-lead compound may have been added.

Red. The red seems to be a mixture of crushed quartz with lead and barium minerals, coloured with a large amount (c. 18%) of hematite powder. However it also possible that the hematite ore itself may have contained abundant quartz. The low alumina and soda contents of the red glaze suggest that the source of quartz may not have been the quartz-feldspar material typical of the blue and white glazes. In addition, the BaO/(BaO+PbO) ratio is significantly lower than in the other glaze colours, at 0.27. Thus the raw materials for the lead and silica of the red may have differed from those of the other glazes

Yellow. The absence of crushed quartz and albite, together with higher levels of alumina, lime and magnesia in this glaze, suggest that its iron content may have been supplied by a fusible ferruginous clay. Indeed a mixture of a red clay with a lead-barium ore, in about 1:1 proportions, would explain both the chemistry and the microstructure of this glaze. The above-average alumina level (8.58% Al_2O_3) may have encouraged the growth of celsian during firing, which has rendered the yellow glaze somewhat opaque. This distinguishes the yellow glaze from true Chinese lead-baria glass where the usual opacifier is barium disilicate – most Chinese glasses being too low in alumina to generate barium feldspar (Brill et al 1991).

The high PbO+BaO total (55.1%) in the yellow glaze would have helped to create a solution colour from the iron present in the glaze, and the glaze is visibly more fusible than the white, the blue and the red. This may explain why it was used as a topmost detail on the bead as it unlikely that it could have supported applied glazes without dissolving them. Of all the glazes on the bead, its composition seems closest to later Han lead glazes, such as an example of an Eastern Han green glaze on a vessel in the Victoria and Albert Museum, London (Wood et al 1992):

Table 4 – Warring States yellow bead-glaze and Eastern Han green lead glaze compared

	SiO_2	TiO_2	Al_2O_3	Fe_2O_3	CuO	CaO	MgO	K_2O	Na_2O	BaO	PbO	SO_3	SnO_2
Bead yellow	27.0	0.4	8.6	4.7	<0.1	1.5	0.7	0.7	0.8	20.1	35.0	<0.1	--
Han Green	33.4	0.6	3.9	2.0	3.0	2.0	0.7	0.5	0.4	7.7	43.5	0.6	1.2

It has been suggested that the Eastern Han green glaze containing baria may have been made from a mixture of sandy loess with lead and barium ores, and coloured by a copper-tin mixture, perhaps oxidised bronze (ibid, 1992). Replications of this Han green glaze matured just above 1000°C and from this we might propose a firing temperature for the bead somewhere between 1000° and 1050°C.

The similarity of the Han green glaze to the bead's yellow glaze suggests that Han lead glaze technology could have owed something to the compositional approaches developed and adopted for making the more fusible and translucent types of faience-bodied bead-glazes in the Warring States period.

Relationship to Chinese glass. The analyses show that these glazes are not quite the same as the glasses used in contemporary Chinese glass beads, although there are some interesting parallels.

Table 5 – Analyses of Chinese glass beads (5th-1st C. BC) (1-4 from Shi et al, 1991; 5-7 from An 1996)

	SiO_2	TiO_2	Al_2O_3	Fe_2O_3	CuO	CaO	MgO	K_2O	Na_2O	BaO	PbO
White glaze	58.4	0.2	4.8	0.5	<0.1	0.7	0.4	1.0	2.6	11.2	19.1
Blue glaze	49.9	0.9	4.4	0.5	2.4	0.9	0.4	1.2	2.2	14.8	21.9
Yellow glaze	27.0	0.4	8.6	4.7	<0.1	1.5	0.7	0.7	0.8	20.1	35.0
Red glaze	42.3	<0.1	2.0	17.8	<0.1	2.2	0.4	0.7	1.1	8.9	24.2
1. Blue bead	41.4	0.01	0.89	0.27	2.07	1.37	0.16	0.16	5.94	9.7	37.4
2. Black bead	37.3	0.05	1.19	7.35	0.42	1.89	0.61	0.37	3.75	9.4	37.5
3. Colourless bead	51.3	0.01	0.46	0.1	0.01	0.37	1.5	0.08	6.12	11.4	28.3
4. Blue bead	52.4	0.03	1.21	0.28	1.31	1.48	2.62	0.16	10.1	11.1	19.2
5. Glass eye bead	43.41	-	4.80	0.14	-	0.69	-	0.27	7.28	14.37	26.88
6. Glass eye bead	37.56	-	4.95	0.49	-	1.75	1.41	-	4.47	16.07	32.00
7. Glass eye bead	36.53	-	7.36	0.25	-	1.69	0.39	-	4.73	15.49	31.53

Both Warring States bead-glasses and Warring States bead-glazes are lead-barium silicates with sodium oxide present as a supplementary flux. However the glass-bead analyses are noticeably higher in soda than the glazes, and commonly lower in alumina. SEM analysis shows that most of the alumina and sodium oxide present in the white and blue glazes is in the form of the water-insoluble material albite, which has a theoretical Na_2O content of 11.8% and an Al_2O_3 content of 19.4%. By contrast the glasses in rows 1-4 of Table 4 have sodium oxide contents that are from twice to nine times their alumina levels. Such high soda and low alumina levels in the glasses could not have been supplied by albite and must therefore represent the use of some water-soluble sources of sodium oxide in the original glass batches.

However, an apparently unusual group of glasses exists which is very similar to the bead glazes (see rows 5-7). These are analyses of eye-beads from Guweicun Tomb 1, Huixian, Henan, presented by An Jiayao (1996) and dated to the Middle and Late Warring States Period. These glasses not only have high alumina levels like the bead glazes but their barium/lead ratios are also similar. These connections raise the possibility that they may have been made from similar materials and recipes – and may even have derived from the same locality. The main differences lie in their soda contents, with the soda levels in the glasses being double those of the glazes. If production of these glass eye-beads and glazed-faience were related then the addition of a few percent of a soda-rich material to the bead glazes, perhaps combined with a higher firing temperature, would have produced a workable glass. Conversely the glazes could represent an underfired intermediate material from the glass-making process.

In terms of practical recipes, glass-batch production copes easily with soluble materials, but insoluble ingredients in glazes are a great advantage. While glazes can be fritted to overcome material solubility, there seems some benefit in the use of unfritted glazes in the case of the

glazed bead technology, as unmelted batch material encouraged opacity in the fired glazes, and also supplied a useful stability in firing.

Warring States glazed vessels. Perhaps the best parallels to the glazed-bead technology described above can be seen in the small group (four known examples) of Warring States pottery vessels, decorated with red-brown, yellow, blue and white glazes and bearing patterns closely related to those seen on the Chinese faience beads.

The authors studied one highly weathered example in the British Museum's collection, semi-quantitatively in 1995 (Wood and Freestone, 1995). As with the bead glazes, lead oxide and baria were the main fluxes, and the glaze-colours were yellow and red from ferric oxide, and blue from effenbergerite. The vessel's body, however, was thrown from ordinary gritty red earthenware clay.

Fig. 5. Chinese Warring States pottery jar with lead-baria coloured glazes. Height 9.5 cm. OA 1968 4-22.18. British Museum Photograph by Tony Milton.

Conclusion

Eye-beads beads imported to China during the Warring States period represented sophisticated Near Eastern glaze and glass technologies that exploited such silicate-colours as cuprite-red, cobalt blue, copper-turquoise, calcium antimonate-white and lead-antimonate yellow (Kaczmarczyk and Hedges 1983, Bimson and Freestone 1987, Lilyquist and Brill

1993). Eye beads using some or all of these colours were manufactured at various centres in the Near East, the eastern Mediterranean, and also at some sites in Western Asia (Yoshimizu 1989).

In China two technologies were employed to copy these imports – true lead-barium glasses, and lead-barium glazed ceramic beads. Our analyses of one example of the latter type have shown an apparently unparalleled body-type made entirely from fused plant ash, and ceramic glazes made from crushed siliceous materials, combined with lead and barium oxides. The bright colours of the Western beads were imitated on this glazed ceramic bead with a more subdued palette that included an iron-yellow, an iron-red and a copper-barium-silicate blue.

The faience-bead glazes differ from contemporary Chinese glasses in some important respects, most notably in their use of insoluble and unfritted ingredients. In some cases these various glaze colours were superimposed in four layers – an effect made possible by good glaze-viscosity and moderate underfiring. This unusual technology seems to have been extended to include the production of small group of glazed thrown vessels, although in this case the faience body was replaced with ordinary earthenware clay (Wood and Freestone, 1995).

Taken together, the glazed bead and the decorated pottery vessel seem to show a very early use of low-fired glazes in China, an early use of copper as a glaze colourant in China, the first use of iron-red and iron-yellow glazes in China, and the first use of polychrome glaze-decoration in Chinese ceramics. In their manipulation of colour, base-glaze composition, and glaze-viscosity, they represent a more complicated and sophisticated technology than that used for contemporary clay-and-ash stoneware glazes in southern China (Zhang 1986b). Nonetheless, the making of Chinese glazed-faience ended in the late first millennium B.C., and it was not until lead-based overglaze enamels were developed in the country in the late 12th century AD that certain aspects of this technology were rediscovered, particularly the use of the iron-red earthenware glaze. Use of the effenbergerite-blue in Chinese glazes, however, seems to have been restricted to the early period, and was not revived.

As to the larger question of how such an unusual and distinctive approach to glass- and glaze making arose in China in the Warring States period, this remains a problem. The importance of lead in these technologies may suggest some connection with metallurgy, but pigment-production may have been another route to lead-rich glasses and glazes. At present a proper chronology for glass, glazed faience and barium-copper-silicate pigment technologies is lacking for China, but this aspect of the subject should become clearer with further excavations and investigations.

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[1.] *Effenbergerite*. This copper barium tetrasilicate was first identified chemically by Pabst in the 1950s; Pabst also discovered the synthetic version of the mineral in early Chinese paint pigments before 1981 (Pabst 1959, FitzHugh and Zycherman, 1992). Giester and Rieck (Giester and Rieck, 1994) found a natural form of the mineral in South Africa and named it effenbergerite after Dr Herta S. Effenberger, the noted authority on divalent copper.

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战国时期中国陶瓷琉璃珠上的早期多彩釉

Early Polychrome Glazes on a Chinese Ceramic Bead of the Warring States Period

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Abstract

A Chinese glazed bead with a faience body, from the Warring States period, was microsampled and studied by scanning electron microscopy, by EDXRF analysis, and by X-ray diffraction. The red, white, blue and yellow glazes on the bead's surface were analysed, as was the pale grey vitreous body beneath the glazes.

The amorphous and vesicular body of the bead gave an analysis characteristic of ash from plants with quick-growing and tough siliceous stems, such as bracken, rice-straw or sorghum. Such ashes are high in silica, potassia, magnesia, calcia and phosphorous oxide. No other ingredients appear to have been used in the bead's body-which seems to be a rare example of a ceramic body made entirely from plant ash.

The coloured glazes on the bead's surface were applied to produce a grid pattern with dotted details and were often superimposed, in one case to four layers. They were not fully melted and contained relict batch material-mainly quartz and albite. They were therefore unfritted and seem to have been fired to earthenware temperatures. These glazes were essentially lead-oxide/baria silicates, sometimes supplemented with soda feldspar, and similar in some ways to early Chinese glass. The glaze colours were: RED-from ferric oxide in suspension; YELLOW-from ferric oxide in solution; WHITE-largely opacified with quartz; and BLUE-like the white but coloured with the copper-barium-tetrasilicate, effenbergerite (Chinese blue). It is not known if the Chinese blue was a product of the firing (i.e. from copper in a barium-rich silicate groundmass) or whether it had been added to the glaze as a preformed blue 'ceramic stain'.

Visually the bead appears to be a Chinese version of imported Western 'eye-beads'. However the composition of its body and glaze are totally Chinese in their technologies and seem related variously to Chinese paint-pigments, Chinese glass technology, and to the so-called 'glass-paste decorated' ceramic vessels of the Warring States-recently studied by the present authors. The use of plant ash as a ceramic body seems to have neither precedents nor descendants in

ceramic history.

引　论

战国早期(公元前480～公元前221年)的中国玻璃质的蜻蜓眼琉璃珠被认为是第一批制造的玻璃人工制品的代表[1,12]。其式样风格被认为源于进口的西方类似于装饰品的琉璃珠，但由于它们的氧化铅-氧化钡-二氧化硅的组成特征又被看成是中国的，这类组成国外是很少有的[1,4]。然而与早期的中国玻璃传统相类似，可以发现一些特有的中国施釉陶瓷琉璃珠往往是制成"蜻蜓眼琉璃珠"的式样。这些珠具有硬陶式胎体和施加颜色釉，设置的图案花纹类似于完全由玻璃制作的蜻蜓眼的图案。

在中国硬陶器是相当少的，但是本文主题所讨论的多彩釉琉璃珠却有十分像硬陶的外观。其胎体组成为硬质的奶白材料，覆盖一层乳浊白釉，施加在白色底釉上的是赭红色、乳黄色和冷蓝色的釉，往往明显地显示出正规的几何图案。

具有低温釉的多彩釉硬陶琉璃珠在中国的西周早期(公元前1027～公元前771年)曾发现，但一般认为是从西方进口的[20]。虽然Brill，Tong和Zhang[3]曾发现有一种琉璃珠的间隙玻璃中是富含钾，也许这与来源于中国是一致的，而后来的出版物[5]考虑了琉璃开始的来源，从而在讨论这里描述的中国最早的低温釉的技术成为可能。他们带有这个国家使用的最早的低温多彩釉的实例也更加可能。

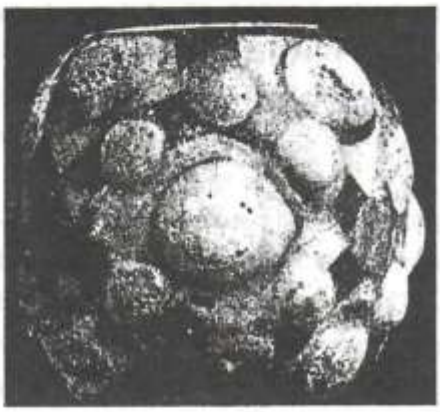

图1. 中国战国时期的硬陶琉璃珠，在白色基釉上有褐红色、黄色、蓝色和白色釉

Fig. 1 Chinese Warring States faience bead with red-brown, yellow, blue and white glazes on a white glazed ground. L. 2 cm, analysed in the present study

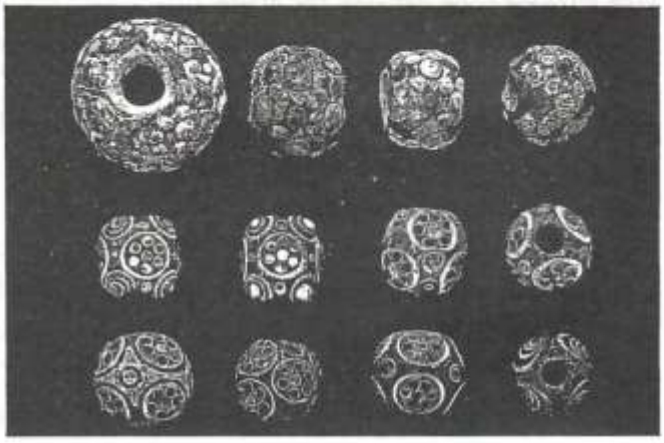

图2. 典型的中国战国时期硬陶蜻蜓眼琉璃珠(顶上一排)和玻璃蜻蜓眼琉璃珠(第二和第三排)[10]

Fig. 2 Typical Chinese Warring States faience eye beads (top row) and glass eye beads (second and third rows)[10]

尽管许多中国玻璃琉璃珠的实例已经分析过，它们与同时期的施釉的陶瓷琉璃珠的关系还不清楚。为了弄清这些早期中国的制作陶瓷琉璃珠使用的工艺和了解它们与当时中国玻璃的联系，我们曾分析了1粒单一的中国施多彩釉的战国时期的琉璃珠，这粒琉璃珠是非常有意义的，因为在式样上和技术工艺上其釉呈现出与1件战国时期的土陶罐有密切关系。土陶罐曾由

Wood 和 Freestone 于 1995 年报告过。本研究所使用的主要技术为扫描电镜和能量散射 X-射线分析(SEM),带有 X-射线衍射(XRD),用来确认若干矿物的一致特征。

结　果

琉璃珠胎体

按 SEM 分析结果,琉璃珠的灰色石质胎体具有很高的非均质结构,由几种富玻璃质的碎片所组成,它们被烧结在一起,但并未全部熔融和流动,留有很高的开口气孔的比例。这些玻璃颗粒在组成上变化很大,在某些情况下有部分失透而沉析出的晶体,长石和磷灰石。全分析表明约有70%的二氧化硅——许多粘土或玻璃所特有的,但是远较西亚来的硬陶低。然而胎体既非富铝像粘土那样,又非富碱像玻璃那样,而是富碱土和富磷质的,具体组成如表 1 所示。

表 1　琉璃珠胎体的全分析

Table 1　Analysis of bulk body of stonepaste bead

	SiO_2	Al_2O_3	Fe_2O_3	CaO	MgO	K_2O	Na_2O	P_2O_5	MnO
%	69.4	3.5	1.4	7.8	5.7	5.0	1.8	4.7	0.3

尽管在外貌上表面近似于西方的硬质陶,玻璃珠的胎并非由大量石英组成,它乃是如同正常确定的硬质陶的基本组成。

琉璃釉

琉璃珠上的釉含有许多矿物包裹物(常常为石英和钠长石)悬浮在氧化铅和富钡的玻璃中。多角形颗粒形状与长石上的反应边缘相耦合而成,表明它们是釉料烧成的遗留物,而非失透后的产物。4 种有色的组成如表 2 所示。

表 2　琉璃珠上所用的白、蓝、黄和红色的釉

Table 2　White, blue, yellow and red glazes used on stonepaste bead

	SiO_2	Al_2O_3	TiO_2	Fe_2O_3	CuO	CaO	MgO	K_2O	Na_2O	BaO	PbO	Cl
白色	58.4	4.8.	0.2	0.5	<0.1	0.7	0.4	1.0	2.6	11.2	19.1	0.8
蓝色	49.9	4.4	0.9	0.5	2.4	0.9	0.4	1.2	2.2	14.8	21.9	0.5
黄色	27.0	8.6	0.4	4.7	<0.1	1.5	0.7	0.7	0.8	20.1	35.0	0.5
红色	42.3	2.0	<0.1	17.8	<0.1	2.0	0.4	0.7	1.1	8.9	24.2	0.5

按其结构而论,各种釉表示有如下的特性:

乳白釉:富有石英的多角片,达到约为 0.2mm 的直径,与其次量的大小相近的钠长石一起乳浊此釉,玻璃基质没有达到平衡,而且铅的分布是不均匀的。低含量的 Fe_2O_3、CaO 和 MgO 提示它们是与主要成分有关联的杂质而非有意的添加物。

乳蓝釉:该釉也含有很多部分反应的石英和钠长石颗粒。另有小晶体长度<0.005mm,确证为硅酸钡铜,含有大约 BaO 31%,CuO 15%和 SiO_2 51%,非常接近稀有矿物 $BaCuSi_4O_{10}$的理想组成。该矿物的存在是由 XRD 分析确证的。晶体的形态学提示它们是从釉的混合物中生长出来的。因此它们是与该矿物的合成变体有关,汉蓝。

乳浊红褐色釉：石英颗粒存在于红褐色釉中，但不像在白釉中那样多。釉的浅红色及其甚乳浊是来自磁铁矿的多角颗粒，直径大约 0.07mm，没有发现长石包裹物，这反映出在釉中的低氧化铅水平(1.96%)。釉施加得很薄，与黄釉 0.3mm 相比在断面上仅达 0.1mm。

乳黄釉：釉中有少量的石英颗粒，其乳浊度主要是由于富含钡长石 $BaAl_2Si_2O_8$，钡长石的晶体生长得很好(XRD 确证)，其晶体尺寸长度<0.005mm。在釉中的氧化铁多数呈玻璃相存在(浓度为 4.6%)，估计是 Fe^{2+}，它可提供黄颜色。氧化铝，9%已是相当高，氧化铅和氧化镁也比较高(分别为 1.5%和 0.74%)，存在于蓝釉和白釉中。

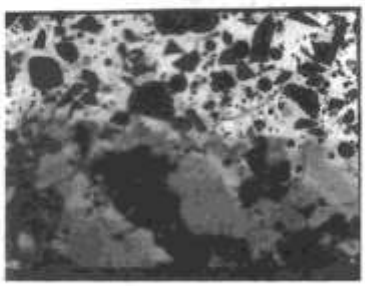

图 3 SEM 背散射成象显示出无定形多孔状琉璃胎体和含多角石英和钠长石碎片在高 Pb+Ba 玻璃中的白色釉

Fig. 3 SEM back-scattered image showing amorphous vesicular bead-body and white glaze containing angular quartz and albite fragments in a high Pb+Ba glass

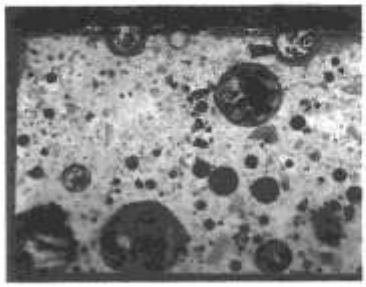

图 4 呈现大而多孔状的黄釉的 SEM 背散射成象。缺少多角状片和细的第二钡长石晶体。同样尺寸的显微照相，大的气泡在图 4 中顶部中心，直径约为 100 mm

Fig. 4 SEM back-scattered image of yellow glaze showing large vesicles, lack of angular fragments and fine secondary celsian crystals

讨　论

琉璃珠胎体

琉璃珠胎体似乎既非玻璃质又非粘土质，其组成类似于快速生长具有坚硬硅质茎干植物的灰，如稻草、蕨类植物(凤尾草之类)、高粱等其组成如表 3 所示。

表 3 快速生长硅质茎干植物的灰[21]

Table 3 Ashes of plants with fast-growing siliceous stems [21]

	SiO_2	Al_2O_3	Fe_2O_3	CaO	MgO	K_2O	Na_2O	P_2O_5	MnO
整体琉璃珠胎	69.4	3.5	1.4	7.8	5.7	5.0	1.8	4.7	0.3
稻草灰	75.8	2.1	0.5	5.8	2.6	6.7	1.45	4.7	0.6
蕨类植物灰	70.8	5.5	1.02	8.5	4.0	5.2	0.11	1.2	1.8
高粱灰	70.8	5.5	2.5	7.6	3.8	6.0	0.6	1.6	1.8

这些比较增加了可能性，即硬质陶胎完全是由植物灰制作而成，高含磷量特别表明是植物灰的使用，正如其原料的不均一性和高易熔性一样。如果胎是由灰制成，则若干有机的粘合剂和浆

料曾用来帮助琉璃珠的成型，把灰粘合在一起。这可以在一定程度上解释其高可能性。一种陶瓷胎完全用植物灰制作是不常有的，本证明似乎在陶瓷史上既无先例又无后继派生事例。无定形材料覆盖在石英为基的多晶硬质陶成分上的优点是与铅钡釉更协调一致的扩展[13]。这类灰很好的烧结特性也提供了一种比基于纯石英的硬质陶更坚硬的和有韧性的胎体。

琉璃珠釉

尽管有些生疏，施在琉璃珠胎上的颜色表面实在全是釉。在应用之前是没曾熔块化过的，其中一些可在 SEM 照相上看出它们全是原始配方材料。然而，不完全的熔化证明在这些釉中可能需要保存施加和添加的彩釉图案花纹的完整性，以及保持其乳浊性和颜色。如这些全部熔化，则会使之混并、流动以及丧失其清晰度。这种层状的方法制作的彩色装饰比同时期的中国玻璃蜻蜓眼琉璃珠有更高的轮廓浮雕感。玻璃蜻蜓眼乃是将不同彩色玻璃借火焰加工和乳光化方法制成同一水平高度的装饰。按照釉的原始组成，以下的方法似乎是可能的。

白色

白色釉由粉碎的石英和钠长石与铅和钡等材料混合而制成。进行小试样面积的分析，在白色、蓝色和黄色釉处的 BaO/(BaO+PbO)比例是相对稳定的，分别为 0.36，0.4 和 0.36。自然产生的铅一钡矿物显然是不清楚的，但具有相对恒定铅钡比的矿物储藏是可以利用的：可以用于这种釉的最简单的方法就是将粉碎的石英-长石岩，如伟晶花岗岩或细晶岩与粉碎的铅钡矿按 2:1 的比例混合。然而使用分别加入石英、长石、铅矿和钡矿制作的方法也不能排除。

蓝色

具有铜的添加物的白釉。铜可能是以氧化金属添加，或许是以天然铜矿物添加。然而它可能是以制成的中国蓝的方式加入的。中国蓝是铜加钡的四硅酸盐，用少量的氧化铅和同时用其紫色变体制备之，在中国的战国和汉代常常作为着色颜料使用[7,8,10]。虽然釉中发现的汉代蓝色似乎是烧成时的结晶，但这些结晶从溶解的中国蓝添加物中发展和生长出来是不可能的。其可能性乃是受到在蓝釉中钡铅比稍高于在黄釉和白釉中(见以上)的情况的支持。这暗示着可能是添加了富钡无铅的化合物。

红色

红色似乎是一种粉碎的石英和铅钡矿物的混合物，用大量的磁铁矿粉(约 18%)着色。然而磁铁矿本身含有丰富的石英是不可能的。红釉的低铝和钠含量暗示石英的来源可能不是典型的蓝釉和白釉的石英-长石原料。加之 BaO/(BaO+PbO)比其他釉彩低得多，为 0.27。所以红色的铅和硅的原料可能不同于其他釉的原料。

黄色

没有粉碎的石英和钠长石，釉中有较高的氧化铝、氧化钙和氧化镁提示其含铁量是由易熔含铁高的粘土提供。的确一种红粘土与一种铅钡矿按 1:1 的比例混合，可既在化学上又可在釉的结构上作出解释。高于平均铝量水平(8.58%Al_2O_3)可以促进在烧成时钡长石的生长. 它可以使得黄釉有些乳浊，这点是黄釉区别于中国的铅钡玻璃的。铅钡玻璃常用的乳浊剂是硅酸钡，大多数中国玻璃是颇低的氧化铝产生钡长石[4]。

黄釉中高的 PbO+BaO 总量(55.1%)会帮助形成一种溶液颜色，由于釉中铁的存在，釉比白色、蓝色和红色釉更明显易熔，这点可以解释为什么它用于琉璃珠上的最上面的图案花纹细节是靠不住的，它可以提供施加的各种釉溶解它们的证据。在琉璃珠上所有的釉，其组成似乎最接近汉代晚期的铅釉，如同伦敦的 Victoria&Albert 博物馆里的容器上东汉绿釉的一个例子[18]。

表 4　战国时期黄色琉璃珠釉和东汉绿铅釉的比较

Table 4　Warring States yellow bead-glaze and Eastern Han green lead glaze compared

	SiO_2	TiO_2	Al_2O_3	Fe_2O_3	CuO	CaO	MgO	K_2O	Na_2O	BaO	PbO	SO_3	SnO_2
琉璃珠黄色	27.0	0.4	8.6	4.7	<0.1	1.5	0.7	0.7	0.8	20.1	35.0	<0.1	—
东汉绿色	33.4	0.6	3.9	2.0	3.0	2.0	0.7	0.5	0.4	7.7	43.5	0.6	1.2

东汉绿釉含钡提示了它可能是由一种含砂黄土与铅、钡矿的混合物制成，并以铜锡混合物，可能是氧化了的青铜着色而成[18]。汉绿釉的复制正好在1000℃以上，由此我们可以得出琉璃珠的烧成温度是在1000℃至1050℃之间的某个温度。

汉带绿釉与琉璃珠黄釉的相似性提示了汉代铅釉的工艺技术可能是靠着一些战国时期制作易熔和透明硬胎体琉璃珠釉发展和采用的组成的方式方法。

与中国玻璃的关系

分析表明这些釉不完全与同时期中国玻璃质琉璃珠用的玻璃相同，虽然有若干有趣的类似之处，如表5所示。

表 5　中国玻璃琉璃珠的分析(B. C. 5世纪～1世纪)[1,18]

Table 5　Analyses of Chinese glass beads (5th-1st C. B. C.)[1,18]

	SiO_2	TiO_2	Al_2O_3	Fe_2O_3	CuO	CaO	MgO	K_2O	Na_2O	BaO	PbO
白釉	58.4	0.2	4.8	0.5	<0.1	0.7	0.4	1.0	2.6	11.2	19.1
蓝釉	49.9	0.9	4.4	0.5	2.4	0.9	0.4	1.2	2.2	14.8	21.9
黄釉	27.0	0.4	8.6	4.7	<0.1	1.5	0.7	0.7	0.8	20.1	35.0
红釉	42.3	<0.1	2.0	17.8	<0.1	2.2	0.4	0.7	1.1	8.9	24.2
1 蓝色玻璃珠	41.4	0.01	0.89	0.27	2.07	1.37	0.16	0.16	5.94	9.7	37.4
2 黑色琉璃珠	37.3	0.05	1.19	7.35	0.42	1.89	0.61	0.37	3.75	9.4	37.5
3 无色琉璃珠	51.3	0.01	0.46	0.1	0.01	0.37	1.5	0.08	6.12	11.4	28.3
4 蓝色琉璃珠	52.4	0.03	1.21	0.28	1.31	1.48	2.62	0.16	10.1	11.1	19.2
5 玻璃蜻蜓眼	43.41		4.80	0.14		0.69		0.27	7.28	14.37	26.88
6 玻璃蜻蜓眼	37.56		4.95	0.49		1.75	1.41		4.47	16.07	32.00
7 玻璃蜻蜓眼	36.53		7.36	0.25		1.69	0.39		4.73	15.49	31.53

无论是战国琉璃珠玻璃还是战国琉璃珠釉其铅钡硅酸盐和氧化钠都是呈现为辅助熔剂。然而玻璃琉璃珠的分析其苏打含量明显高于釉，而铝含量通常较低。SEM分析表明，很多铝和钠的氧化物在白色和蓝色中是呈不溶解的钠长石原料存在着，其理论Na_2O含量为11.8%而Al_2O_3位19.4%。相反表5中1～4的玻璃中的氧化钠含量是含铝量的2～9倍。玻璃中如此高的钠和低的铝含量，可能不是钠长石所提供，肯定是在原始玻璃配方中使用了一些可溶性的氧化钠原料。

然而一组显然不常有的玻璃组存在着，它们与琉璃釉非常相似(见表5中的5～7)，有河南Hui县Guwei村墓葬出土的蜻蜓眼琉璃珠的分析1[1]，时代为战国中期和后期。这些玻璃不仅有高含铅量，如同琉璃珠釉一样，而且其钡/铅比也相近。这些联系增加了它们是用近似材料和方法制作的可能性，甚至可能是出自同一地区。主要的差别是它们的苏打含量的不同，玻璃的苏打含

量为釉的 2 倍。如果这些玻璃蜻蜓眼的生产制作和上釉硬质陶是有关系的，那么多加百分之几的富苏打原料到琉璃珠胎中也许与高的烧成温度结合起来就会制做出容易加工的玻璃。相反这些釉也可能是来自玻璃制作过程的生烧的中间材料。

按照实际方法，玻璃料方的制作很容易应用可溶性原料，但釉中不溶的成分是一个很大的有利条件，当釉被熔块化超过材料的溶解度时，似乎在施釉琉璃球工艺的情况中使用未熔块化的釉有些益处。因为不熔化的配料原料促进在烧成釉中的乳浊度，并提供烧成中有用的稳定性。

战国时代上釉器皿

也许与上述施釉琉璃珠工艺最好的对比能在战国陶器容器的小组（4 个已知的例子）中看出来。陶器容器是用红褐色、蓝色、黄色和白色釉装饰的，显示出的图案花纹与在中国硬质陶琉璃珠上的花纹有密切关系。作者们研究了一个大英博物馆收藏的高度风化的粒子，半定量的研究进行于 1995 年[17]。各种琉璃珠釉氧化铅和氧化钡是主要熔剂，而釉彩的黄色和红色是来自氧化铁，蓝色是来自 $BaCuSi_4O_{10}$矿物（effenbergerite）。然而容器的胎则是用普通的砂质红陶器粘土制作。

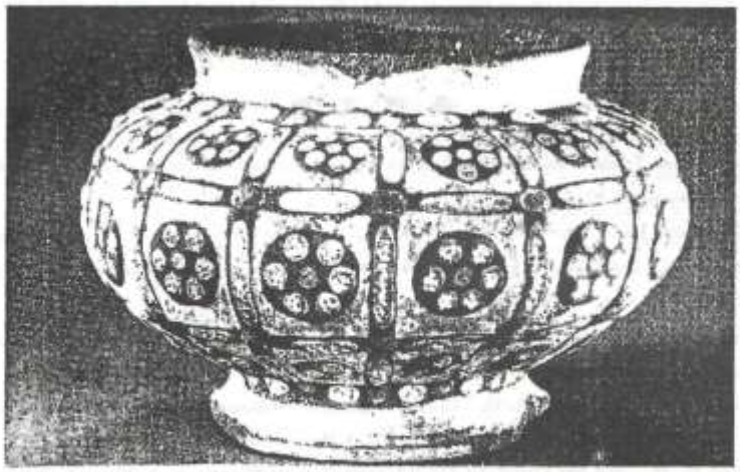

图 5　中国战国时期以铅钡彩色釉装饰的陶罐，高度 9.5 cm。
OA 1968 4-22.18 大英博物馆 Tony Milton 照相
Fig. 5　Chinese Warring States pottery jar with lead-baria coloured glazes. Height 9.5 cm. OA 1968 4-22.18

结　论

中国战国时期仿制的蜻蜓眼琉璃珠表现出精致的近东釉和玻璃的工艺技术，开拓出了一些硅酸盐彩料，如赤铜矿红色、钴蓝色、铜一松绿石色、锑酸钙白色、锑酸铅黄色[2,11,13]。蜻蜓眼琉璃珠使用若干或全部颜色是在近东各中心、东地中海和西亚的一些城市制造的[19]。

在中国有两种工艺技术用来复制这些进口品——真正的铅一钡玻璃和铅钡釉的陶瓷琉璃珠。我们对后面一种形式的一例分析表明一种明显无前例的胎的型式是完全由熔融植物灰制成，而且陶瓷釉是由粉碎硅质材料与铅钡氧化物相结合制成。西方琉璃珠的明亮色彩也在这种上釉的陶瓷琉璃珠上仿制呈一套更柔和的颜色，其中包括一种铁黄、一种铁红和一种硅酸铜钡蓝色。

硬质陶琉璃珠釉在若干重要方面是不同于同时期的中国玻璃的。最主要的是它们使用不溶

解的和未熔块化的成分。在某些情况下这些各种的釉彩呈四层加在上面——其效果可能是由于很好的釉的粘度和适中的生烧所形成。这种特种的技术似乎已经延伸到包括小量的施釉器的生产，虽然在这种情况中硬质陶胎体是换成了普通的土器用粘土[17]。

综合起来，施釉的琉璃珠和彩饰的陶器似乎表明了在中国很早就使用了低温釉、早就使用了铜作为釉的着色剂、在中国第一次使用铁红釉和铁黄釉、第一次使用多彩釉装饰在中国的陶瓷上。在其颜色控制、基釉组成和釉的粘度各方面都显现了比当时中国南方制作粘土灰石器用的釉更复杂的和老练的技术[22]。然而，中国施釉硬质陶的制作是在 B.C. 晚期第一个千年的末尾，而且直至在公元 12 世纪后期在国内发展了铅基釉上彩之前均没有这种施彩釉硬质陶。这些技术的某些方面在 12 世纪之后又被重新发现，特别是铁红土陶器釉的使用。硅酸铜钡蓝釉彩色用于中国釉中似乎在早期受到限制而未得到再现。

至于较大的问题是如此一个特殊的富有特色的釉和玻璃的制作方法在中国的战国时期出现，这仍旧是一个问题。在这些技术中铅的重要性可以提示着是与冶金技术有某些联系，但是颜料的生产可能是导致富铅玻璃和釉的另一个途径。现在一个适当的有关玻璃的、施釉硬质陶的和硅酸钡铜颜料等各种技术的纪年表在中国还没有，但是随着进一步的发掘和研究主题的确立，这些方面将会更加清楚。

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Effenbergerite. This copper barium tetrasilicate was first identified chemically by Pabst in the 1950s; Pabst also discovered the synthetic version of the mineral in early Chinese paint pigments before 1981(Pabst 1959, Fitzhugh and Zycherman, 1992). Giester and Riech (Giester and Riech, 1994) found a natural form of the mineral in South Africa and named it effenbergerite after Dr. Herta S. Effenberger, the noted authority on divalent copper. The mineral was identified in a Chinese glass by Brill et al in 1991(Brill et al 1991), and in a Chinese glaze by Wood and Freestone in 1995(Wood and Freestone 1995).

（郭演仪译）

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